INCH-POUND

MIL-DTL-24416C(SH)
30 March 2006

SUPERSEDING
MIL-E-24416B(SH)
8 February 1991

DETAIL SPECIFICATION

EXTINGUISHING SYSTEM, FIRE, FIXED, WET CHEMICAL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for pre-engineered wet chemical fixed fire extinguishing systems for protection of deep fat fryers, and doughnut fryers onboard U.S. Navy ships and submarines. The systems covered under this specification are intended for use in new applications, and for backfit into existing wet chemical extinguishing system installations. The current wet chemical extinguishing system utilizes aqueous potassium carbonate (APC) extinguishing agent, in a concentration of 42.4 percent K_2CO_3 and 52.8 percent water.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-595	-	Colors Used in Government Procurement

COMMERCIAL ITEM DESCRIPTIONS

A-A-59666	-	Cylinder, Compressed Gas: DOT Specifications 4B, 4BA, 4BW, and 4E

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements For

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05Q, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to commandstandards@navsea.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:52Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

GENERAL SERVICES ADMINISTRATION (GSA)

General Services Administration (GSA) Global Supply Catalog

(Copies of this document are available from the GSA Centralized Mailing List Service (CMLS) by phone at (817) 334-5215, by fax at (817) 334-5561, or online at www.gsa.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASME INTERNATIONAL

ASME Boiler and Pressure Vessel Code Section VIII

ASME B40.100 - Pressure Gauges and Gauge Attachments - Part I - B40.1 Gauges: Pressure Indicating Dial Type: Elastic Element (DoD adopted)

(Copies of these documents are available from ASME Service Center, 22 Law Drive, Box 2900, Fairfield, NJ 07007-2900 or online at www.asme.org.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 17A - Standard for Wet Chemical Extinguishing Systems (DoD adopted)

NFPA 96 - Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations (DoD adopted)

(Copies of these documents are available from NFPA, 1 Batterymarch Park, Quincy, MA 02169-7471 or online at www.nfpa.org.)

NCSL INTERNATIONAL

ANSI/NCSL Z540.1-1994 - Calibration and Measurement and Test Equipment General Requirements (DoD adopted)

(Copies of this document are available from NCSL International, 2995 Wilderness Place, Suite 107, Boulder, CO 80301-5404 or online at www.ncsli.org.)

UNDERWRITERS LABORATORES INC. (UL)

UL 300 - Fire Testing of Fire Extinguishing Systems for Protection of Commercial Cooking Equipment

(Copies of this document are available from COMM 2000, 1414 Brook Drive, Downers Grove, IL 60515 or online at www.ul.com.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:52Z
Check the source to verify that this is the current version before use.

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article test. When specified (see 6.2.b), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material. The material for wet chemical systems and all associated components shall comply with standards UL 300, NFPA 17A, and NFPA 96. When a definite material is not designated, the contractor shall select the best material commercially available that meets the requirements specified herein.

3.2.1 Quality. The material shall be uniform in quality and free from defects that affect performance, serviceability, or appearance, such as lumps, kinks, splits, abrasions, scrapes, corroded surfaces, and impurities.

3.2.2 Combustion and corrosion resistance. Unless otherwise specified, material shall be inherently non-combustible and corrosion resistant or treated to provide protection against heat and corrosion.

3.2.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Environmental considerations.

3.3.1. Operational temperatures. Operating devices shall be designed to function properly through a minimum temperature range from 32 °F to 120 °F (0 °C to 49 °C) or the temperature range specified in the manufacturer's listed installation manual and maintenance manual if the manufacturer's range is greater.

3.3.2 Corrosion. Exposed parts of system installed in the area protected by the extinguishing agent spray shall be constructed of corrosion resistant material.

3.3.3 Shock. The wet chemical system and all associated components shall meet the Grade B shock acceptance requirements of MIL-S-901 when subjected to the test of 4.5.2.

3.3.4 Vibration. The wet chemical system and all associated components shall meet the Type I vibration requirements of MIL-STD-167-1 when subjected to the test of 4.5.3. In addition to the acceptance criteria in MIL-STD-167-1, the pressure switches shall perform as specified in 3.4.6, and the pressure gauges shall be accurate within 4 percent of the true pressure over the central half of the scale range and within 6 percent of the true pressure over the rest of the scale range.

3.4 Design and construction. Wet chemical systems and all associated components shall be listed under UL 300, and shall meet the requirements of NFPA 17A, NFPA 96, and manufacturer's pre-engineered wet chemical system installation and maintenance manuals. The components of the wet chemical extinguishing system include extinguishing agent, extinguishing agent cylinder, expellant gas cylinder (when applicable), cylinder discharge valves, discharge nozzles, remote manual actuation devices, pressure switches, pressure gauges, distribution piping, cable, conduit, pulleys, mounting hardware, fusible links, cylinder actuating mechanism, and an interface for fryer automatic shut-off device. Unless otherwise specified (see 6.2.c), the supplier shall provide the dimensional requirements for system installation and operation required for the wet chemical system to perform its function. Wet chemical system and all associated components shall be accessible for inspection, maintenance, and recharge. The wet chemical system shall be outfitted with a visual indicator that provides operational status of the wet chemical system (i.e., "ready" or "requires recharging"). Systems and components supplied under this specification shall be listed under UL 300 by the manufacturer for use with APC in the concentration of 42.4 percent K2CO3 and 52.8 percent H2O using the criteria specified in the procurement (see 6.2.d).

3.4.1 Extinguishing agent. The wet chemical system shall use APC for the extinguishing agent.

3.4.2 Extinguishing agent cylinder. Extinguishing agent cylinders shall have the ability to be refilled with extinguishing agent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:52Z
Check the source to verify that this is the current version before use.

3.4.2.1 Construction. Cylinders shall be constructed so the pressure gauge and/or visual indicators of operational status can be read.

3.4.2.2 Pressure gauge. A pressure gauge shall be provided for the extinguishing agent cylinder, showing the pressure within the cylinder. The pressure gauge face shall indicate the operating pressure range of the cylinder. Dial pressure gauges shall be in accordance with ASME B40.100. The pressure gauge shall be calibrated and provided with a certificate of calibration in accordance with ANSI/NCSL Z540-1-1994.

3.4.2.3 Cylinder rating. Extinguishing agent cylinders shall be rated for water capacity and service pressure of the wet chemical system. Extinguishing agent cylinders that do not use an external expellant gas cylinder for charging, shall meet the requirements of A-A-59666 when shipping of the cylinder is required in the charged condition. Extinguishing agent cylinders using an external expellant gas cylinder for charging, shall meet the requirements of ASME Boiler and Pressure Vessel Code Section VIII, when cylinders are not shipped in the charged condition.

3.4.2.4 Extinguishing agent discharge valve. Internally pressurized extinguishing agent cylinders shall be outfitted with a discharge valve that controls release of extinguishing agent. Discharge valve shall be designed to provide rapid actuation and discharge of agent. Discharge valve shall be made of corrosion resistant, non-combustible material. The discharge hose or piping shall be made of corrosion resistant, non-combustible material. Discharge valves shall withstand the extinguishing agent cylinder hydrostatic test pressure specified in A-A-59666 or the ASME Boiler and Pressure Vessel Code Section VIII, as applicable, without rupture, damage, or permanent distortion.

3.4.3 Expellant gas. Dry nitrogen or dry air shall be used to facilitate the discharge of the extinguishing agent. Expellant gas may be contained within an internally pressurized extinguishing agent cylinder, or provided through an external self-contained pressurized source integral to the wet chemical extinguishing system.

3.4.4 Expellant gas cylinder assembly. External expellant gas cylinder capacity and pressure ratings shall be as specified in the manufacturer's wet chemical fire extinguishing system installation manual, and shall be provided with a valve assembly to regulate the discharge of expellant gas.

3.4.4.1 Construction. An enclosure or other means shall be provided to protect the expellant gas cylinder from damage. Expellant gas cylinder shall be accessible for inspection, maintenance, and recharge or replacement. The cylinder assembly shall be constructed so that the pressure gauge and/or visual indicators of operational status can be read without removal of the enclosure or guards, and without securing the extinguishing system.

3.4.4.2 Pressure gauge. A pressure gauge shall be provided for the expellant gas cylinder, showing the pressure within the cylinder. The pressure gauge face shall indicate the operating pressure range of the cylinder. Dial pressure gauges shall be in accordance with ASME B40.100. The pressure gauge shall be calibrated and provided with a certificate of calibration in accordance with ANSI/NCSL Z540-1-1994.

3.4.5 Extinguishing agent release mechanism. The wet chemical fire extinguishing system shall have an extinguishing agent release mechanism that activates the extinguishing agent cylinder discharge valve and the external expellant gas cylinder discharge valve (when applicable) to release the extinguishing agent through the piping distribution network. The release mechanism actuation modes shall be automatic actuation by fusible link detectors, manual actuation by remote devices, and manual actuation at the release mechanism.

3.4.6 Pressure switch. Pressure switches shall be provided to operate on low extinguishing agent cylinder pressure (minimum operating pressure) and shall provide an interface to de-energize fryer heating elements and an interface to activate audible and visual alarms if the pressure drops. Pressure switch shall incorporate a manual reset or require recharging of the extinguishing agent cylinder before the pressure switch can be reset. Pressure switch components exposed to extinguishing agent shall be corrosion resistant.

3.4.7 Fusible link. Fusible links shall meet the requirements of NFPA 17A and be in accordance with the wet chemical fire extinguishing system configuration for which NFPA approval was granted.

3.4.8 Vent plug. A device shall be provided to safely relieve pressure and to prevent a slow, unwanted build-up of pressure in the discharge (distribution) piping.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:52Z
Check the source to verify that this is the current version before use.

3.4.9 Discharge nozzles. Discharge nozzles shall be in accordance with the wet chemical fire extinguishing system configuration for which NFPA approval was granted. Discharge nozzles shall be constructed of a non-combustible corrosion resistant material acceptable for use in food service zones.

3.4.9.1 Protective device. Discharge nozzles shall be provided with caps, covers, or other suitable protective devices to protect nozzle from clogging by grease or other obstructions. Protective device shall detach upon actuation.

3.4.9.2 Strainers. Strainers shall be installed at each nozzle to prevent foreign matter inside the distribution piping from clogging the nozzle orifice.

3.4.10 Remote actuation devices. Remote actuation devices shall be provided for the manual actuation of the extinguishing system. Remote actuation devices shall not require a force of more than 40 lbs or a movement of more than 14 inches to operate. Remote actuation devices shall be provided with operating instructions.

3.4.11 Cable sections/actuating cable assembly. A cable and conduit assembly with corner pulleys, or sealed ball bearing control cable are suitable for connection of the system actuation components including, remote manual actuation device, fusible links, and the extinguishing agent release mechanism.

3.4.11.1 Cable. Actuating cable shall be installed enclosed in conduit. Cable shall be of sufficient tensile strength, diameter, and material composition to withstand the pressure of the taut extinguisher release mechanism system which includes fusible link detectors, extinguishing agent release mechanism, and remote manual actuation device. Cable shall not bind at any point in the conduit or corner pulley. Cable conduit shall be installed so that conduit runs cannot be moved out of their proper installed alignment.

3.4.11.2 Corner pulley. Corner pulleys shall be used whenever the cable requires a directional change of 90 degrees. Grease-tight corner pulleys shall be used in areas likely to experience excessive grease deposit build-up.

3.4.11.3 Actuating cable assembly. A sealed ball bearing control cable is acceptable for the connection and actuation of the actuating system components. The ball bearing control cable shall consist of CRES races and guides separated by two rows of CRES balls bearing, sealed in a flexible CRES sheathing. A potted steel cable (both ends) connects the control cable to the actuation components including, remote manual actuation device, fusible links, and the extinguishing agent release mechanism.

3.4.12 Distribution piping and fittings.

3.4.12.1 Material. Distribution piping and fittings shall be made of CRES piping or tubing.

3.4.13 Recharge apparatus. Recharge apparatus, fittings, and adapters shall be provided for recharging extinguishing agent cylinders and expellant gas cylinders.

3.4.14 Color. Cylinders, control boxes, and enclosures shall be painted with red enamel in accordance with color number 11105 of FED-STD-595.

3.4.15 Tools. No special tools shall be required for installation or maintenance of the system. Special tools are defined as those tools not listed in the General Services Administration (GSA) Global Supply Catalog.

3.5 Performance characteristics.

3.5.1 Actuation requirements. Actuation from separation of fusible link detectors or operation of the remote or local manual actuators shall result in discharge of the extinguishing agent.

3.5.2 Extinguishing agent requirements. Extinguishing agent shall perform in accordance with requirements of UL 300 and NFPA 17A. When applied to flammable liquid surfaces, the extinguishing agent shall completely extinguish the fire by creating a foam blanket barrier between the flammable liquid and oxygen and provide cooling to the fat reservoir.

3.5.3 Discharge nozzle requirements. Discharge nozzles shall uniformly distribute the extinguishing agent over and into the protected appliance in accordance with NFPA 17A when tested in accordance with 4.5.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:52Z
Check the source to verify that this is the current version before use.

3.6 Certification by Underwriters Laboratories, Inc. The wet chemical system and all associated component equipment for extinguishment, splash, and discharge (distribution and time) shall be certified in accordance with UL 300 as specified (see 6.2.e).

3.7 Workmanship. The wet chemical system and all associated components shall be uniform in quality and evidence a level of workmanship that conforms to high-grade commercial practice for fire protection equipment. Components shall be free from irregularities, defects, or foreign matter which affect their appearance or which may affect their performance, durability, reliability, serviceability, or safety. Components shall be free of burrs, scratches, chips, sharp edges, loose, or defective parts.

3.8 Special markings. All required markings shall be permanent and as follows:

3.8.1 Extinguishing agent cylinder. Cylinder shall be marked with the following information: Aqueous Potassium Carbonate (name as it appears on MSDS sheet (see 6.3)); operating pressure of unit at room temperature; hydrostatic test pressure and date; inspection instructions; maintenance instructions; and basic cautionary instructions (first aid instructions.).

3.8.2 Discharge nozzle. Discharge nozzles shall be stamped with flow designation and tip part number.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on one complete wet chemical extinguishing system when a first article sample is required (see 3.1). The inspection shall include the examination of 4.4 and the tests of 4.5.1 through 4.5.5 (see Table I).

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4, 4.5.1, and 4.5.5 (see Table I).

4.4 Examination. A visual examination shall be performed on the installed wet chemical system and all associated components. Wet chemical system and all associated components shall be examined for compliance with requirements outlined in 3.2, 3.3.1, 3.3.2, 3.4, 3.6, 3.7, and 3.8 of this specification. This examination shall encompass all visual examinations including compliance verification of system design and installation in accordance with manufacturers UL approved installation and maintenance manuals. Noncompliance with any specified requirements or presence of one or more defects shall constitute cause for rejection.

4.5 Methods of inspection.

4.5.1 Materials test. The requirements of 3.2 shall be verified by certification of conformance to UL 300, NFPA 17A, and NFPA 96 (see 6.2.f).

4.5.2 Shock test. The shock test shall be conducted as follows: A wet chemical extinguishing system incorporating all of the system components required for demonstration of compliance to UL 300 shall be assembled. The assembly shall be affixed to a shock mounting fixture to simulate the worst case mounting location, plane, and orientation for the installed equipment aboard ship. Fresh water may be used as the extinguishing agent charge during the test. The assembly shall be subjected to a Grade B, light weight, Class I, Type A shock test in accordance with MIL-S-901. The assembly shall meet the requirements of 3.3.3 after the test. Any change in system components, parts, or fasteners shall require a retest unless otherwise specified (see 6.2.g).

4.5.3 Vibration test. After the tests of 4.5.2, the wet chemical fire extinguishing system shall be subjected to a Type I vibration test in accordance with MIL-STD-167-1. The system shall not be actuated while being vibrated. After verifying compliance of the pressure switches, the system shall be actuated to demonstrate its ability to perform its function. The wet chemical fire extinguishing system shall meet the requirements of 3.3.4 after the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:52Z
Check the source to verify that this is the current version before use.

4.5.4 Functional test. A functional test shall be performed to demonstrate the reliability and proper functioning of all pressure switches, shut-off devices interface, alarm system interface, visual indicators, mechanical or pneumatic actuating mechanism, remote actuating device, fusible links, cable release system, discharge valves, expellant gas assembly (where installed), and discharge nozzles to meet the requirements of 3.4 and 3.5. Each extinguishing system actuation method shall be individually tested. Failure of installed wet chemical system to meet requirements of 3.4 and 3.5 of this specification shall constitute failure of the functional test.

4.5.5 Hydrostatic test. Conformance to 3.4.2.3 shall be determined by inspection of hydrostatic pressure rating test date in accordance with A-A-59666 or ASME Boiler and Pressure Vessel Code Section VIII as applicable (see 3.4.2.3).

TABLE I. Tests.

Characteristic	Requirement	Test	First article	Conformance
Material	3.2 & 3.3.2	4.4, 4.5.1	X	X
Shock	3.3.3	4.5.2	X	
Vibration	3.3.4	4.5.3	X	
Design and construction	3.3.1, 3.4 – 3.4.2.2 & 3.4.2.4 – 3.4.15	4.4	X	X
Hydrostatic	3.4.2.3	4.5.5	X	X
Performance	3.3.1, 3.4, & 3.5	4.5.4	X	
UL certification	3.6	4.4	X	X
Workmanship	3.7	4.4	X	X
Markings	3.8	4.4	X	X

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2.h). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Pre-engineered wet chemical systems are intended for protection of deep fat fryers and doughnut fryers. A pre-engineered system has predetermined flow rates, nozzle pressures, and quantities of extinguishing agent. These systems have the specific pipe size, maximum and minimum pipe lengths, pie volumes, number of fittings, and number and types of nozzles approved by a testing laboratory.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. When first article is required (see 3.1).

c. When manufacturer's dimensional requirements for system installation and operation are not required (see 3.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:52Z
Check the source to verify that this is the current version before use.

d. Certification of the wet chemical extinguishing system for listing under UL 300 when using aqueous potassium carbonate as the extinguishing agent (see 3.4).

e. Certification of UL 300 listing (see 3.6).

f. Certification of conformance to UL 300, NFPA 17A, and NFPA 96 (see 4.5.1).

g. When shock retest is not required (see 4.5.2).

h. Packaging requirements (see 5.1).

i. Manufacturer's installation, operating, and maintenance manuals (see 6.4).

6.3 Material safety data sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent government mailing addresses for submission of data are listed in FED-STD-313.

6.4 Technical manuals. Manufacturer's installation, operation, and maintenance manuals will be for the configuration delivered (see 6.2.i).

6.5 Subject term (key word) listing.

Aqueous potassium carbonate
Deep fat fryer
Doughnut fryer
Fire protection
Galley equipment
Pre-engineered
Saponification
Wet chemical

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing Activity:
Navy – SH
(Project 4210-0663-000)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:52Z
Check the source to verify that this is the current version before use.